

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
the de Broglie wavelength of a particle is given by the formula $\lambda = h/p$, where λ is the wavelength, h is the Planck's constant, and p is the momentum of the particle.
the momentum of a particle is given by the formula $p = \sqrt{2 * m * E}$, where m is the mass of the particle and E is the kinetic energy.
we are given that the kinetic energy is 1 eV, which is equal to 1.6×10^{-19} J.
the mass of an electron is approximately 9.11×10^{-31} kg.
first, we convert the kinetic energy from eV to Joules: $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$.
the momentum is given by $p = \sqrt{2 * 9.11 \times 10^{-31} \text{ kg} * 1.6 \times 10^{-19} \text{ J}}$.
then, we can calculate the wavelength by dividing the Planck's constant ($6.626 \times 10^{-34} \text{ m}^2 \text{ kg} / \text{s}$) by the momentum.
let's perform the calculation and convert to cm.
after the calculation, we obtain a value of $1.227 \times 10^{-7} \text{ cm}$.
</think><answer> G </answer>

